



Probing Fast Radio Burst Environments

Cross-Correlating Deep and Shallow FRB Surveys with KiDS DR5

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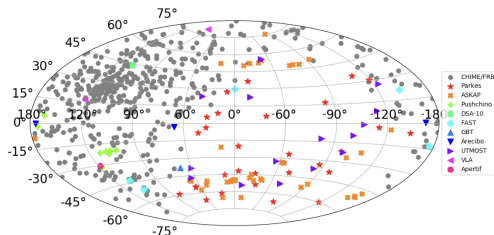
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Motivation

- Fast Radio Bursts (FRBs) are millisecond-duration radio transients of cosmological origin
- Thousands now known (CHIME/FRB: 4539 bursts, 3641 sources), but individual redshifts are rarely measured
- Their true progenitors are still unknown – magnetars vs. delayed neutron-star channels
- **Idea:** cross-correlate the FRB sky distribution with a galaxy survey to statistically infer the FRB redshift distribution and host bias
- **Goal of this thesis:** forecast how well an FRB $\times$ galaxy cross-correlation with KiDS-Legacy can constrain the FRB host-population bias

Fast Radio Bursts

- Bright, ms-duration radio pulses, first discovered in 2007 (Lorimer Burst)
- Dispersion measure $DM \propto \int n_e dl$ reveals cosmological (extragalactic) distances
- Some sources repeat, others appear as one-off events
- Sky distribution consistent with an isotropic, extragalactic population

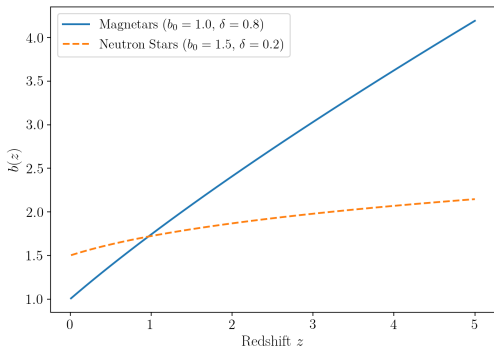


Progenitor Models & Host Bias

- **Magnetars:** young neutron stars from core-collapse supernovae, tracking the cosmic star-formation history
- **Delayed neutron stars:** formed via mergers/AIC, with broad delay-time distributions
- The two classes predict distinct redshift evolutions, encoded as a linear bias

$$b(z) = b_0 (1 + z)^\delta$$

- Distinguishing them is a key open question in FRB astrophysics



Angular Power Spectra & the Limber Approximation

- Clustering of a tracer field is quantified by the angular power spectrum C_{ℓ}^{AB}
- Full projection integral reduces, for $\ell \gg 10$, to the Limber approximation

$$C_{\ell}^{AB} \approx \int \frac{d\chi}{\chi^2} W^A(\chi) W^B(\chi) P\left(k = \frac{\ell + 1/2}{\chi}, z(\chi)\right)$$

- Radial weight functions $W(\chi) = b(z) n(z) H(z)/c$ combine the tracer's redshift distribution and bias
- Auto-spectra ($A = B$) receive an additive Poisson shot-noise term $N_{\ell} = 1/\bar{n}$; cross-spectra are shot-noise free

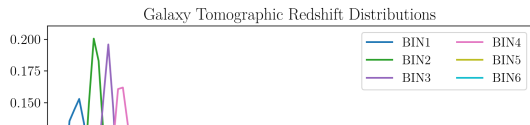
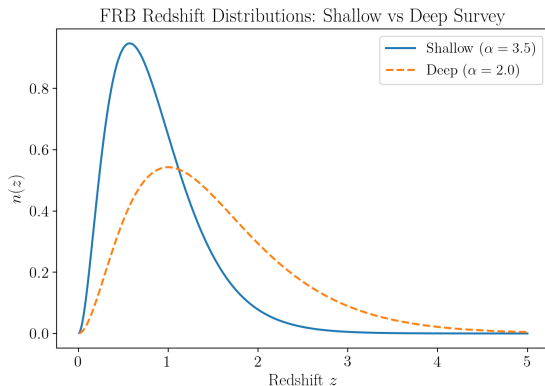
Data & Fiducial Model

FRB mock samples

- $n(z) \propto z^2 e^{-\alpha z}$
- Shallow: $\alpha = 3.5$, $N = 5000$
- Deep: $\alpha = 2.0$, $N = 50\,000$
- $f_{\text{sky}} = 0.9$ for both

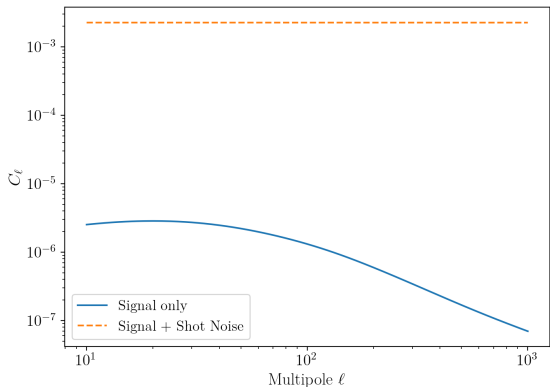
Galaxy sample

- KiDS-Legacy (DR5), 1347 deg^2
- 6 tomographic redshift bins



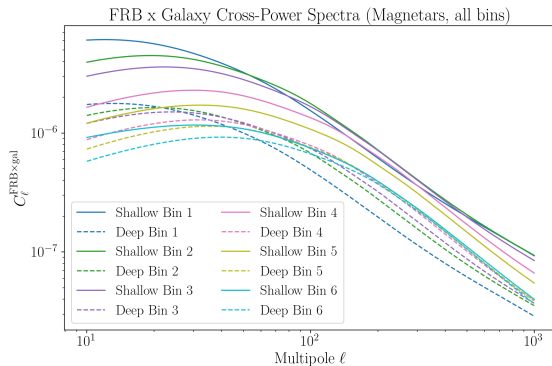
FRB Auto-Correlation is Shot-Noise Dominated

- Planck 2018 flat Λ CDM cosmology, nonlinear $P(k, z)$ from hmf
 - For a shallow sample of $N = 5000$ bursts, shot noise exceeds the clustering signal by ~ 3 orders of magnitude at all multipoles
 - The FRB auto-spectrum alone carries essentially *no* bias information
- ⇒ motivates cross-correlating with a dense galaxy sample



FRB × Galaxy Cross-Spectra

- Cross-spectra computed for both progenitor classes, both survey depths, and all 6 tomographic bins (24 spectra total)
- No shot-noise contamination – signal set by the radial overlap of FRB and galaxy kernels
- Amplitude and shape trace the tomographic bin index and survey depth



Fisher Forecast Formalism

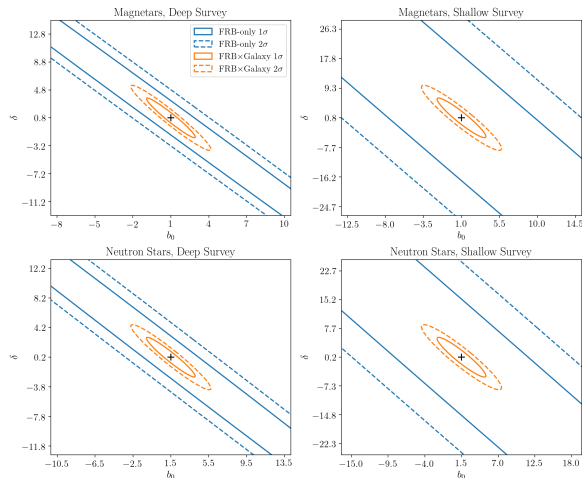
- Gaussian likelihood $\Rightarrow$ parameter covariance from the Fisher matrix

$$F_{ij} = f_{\text{sky}} \sum_{\ell} \frac{2\ell + 1}{2} \text{Tr} \left[C_{\ell}^{-1} \frac{\partial C_{\ell}}{\partial \theta_i} C_{\ell}^{-1} \frac{\partial C_{\ell}}{\partial \theta_j} \right]$$

- Parameters of interest: FRB bias amplitude b_0 and evolution exponent δ
- Marginalized errors $\sigma_i = \sqrt{(F^{-1})_{ii}}$; figure of merit $\text{FoM} = 1/\sqrt{\det \text{Cov}}$
- Compared: FRB-only ($f_{\text{sky}} = 0.9$) vs. joint FRB $\times$ KiDS ($f_{\text{sky}} = 0.0327$)

Forecast Constraints

- FRB-only: $\sigma_{b_0} \approx 23\text{--}101$ (uninformative)
- Joint FRB $\times$ galaxy: $\sigma_{b_0} \approx 1.3\text{--}2.4$
- Improvement: factor **17–51** in σ , factor **61–449** in FoM
- Deep configuration always outperforms shallow, but gains saturate once shot noise no longer dominates

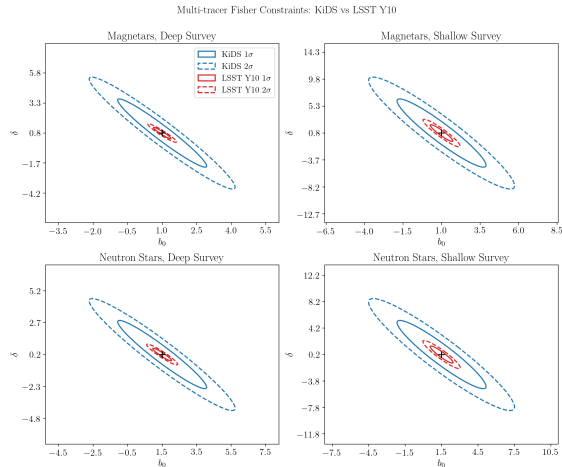


Discussion

- Cross-correlation turns an uninformative FRB auto-spectrum into a competitive bias measurement
- The two progenitor models ($\Delta b_0 = 0.5$, $\Delta \delta = 0.6$) are *not* yet distinguishable: their fiducial separation lies within the 1σ joint contour
- Fisher information scales linearly with $f_{\text{sky}} \Rightarrow$ sky area is the limiting factor, not burst count

Outlook: A Wider Galaxy Survey (LSST)

- LSST Y10 footprint: $18\,000\,\text{deg}^2$, $f_{\text{sky}} = 0.4363$ ($13\times$ KiDS)
- Deep magnetar case: σ_{b_0} improves from $1.27 \rightarrow 0.25$, FoM $\times 26$
- The two progenitor models become distinguishable at 1σ



Conclusion

- Built a full forecasting pipeline: matter power spectrum $\rightarrow$ radial kernels $\rightarrow$ Limber $C_\ell \rightarrow$ Fisher forecast
- FRB auto-correlation alone is shot-noise dominated and uninformative about host bias
- Cross-correlating with KiDS-Legacy improves constraints by 1–2 orders of magnitude
- Current footprint is not yet sufficient to separate magnetar vs. neutron-star origin – but a wider survey such as LSST would be
- FRB $\times$ galaxy cross-correlation is a promising probe of FRB host populations

Thank You

Thank you for your attention!

Questions?